Laboratory for Scientific Computing
UNIVERSITY OF CAMBRIDGE | Cavendish Laboratory
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Mathematical Methods for Scientific Computing

Introduction and Vector Calculus

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Mathematical methods for scientific computing

- Scientific computing could be considered the field of turning equations into code, to solve problems which are otherwise intractable

- We start with a system of equations, e.g. the compressible Euler equations

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$$

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} + Ip) = 0$$

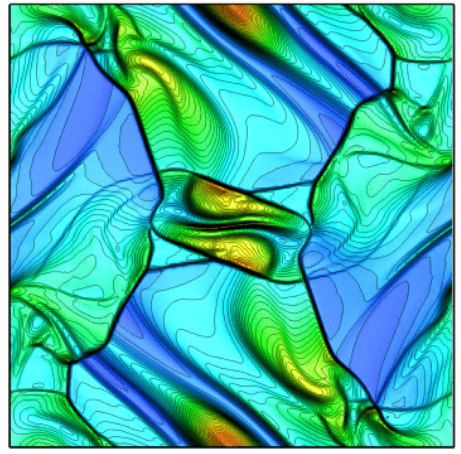
$$\frac{\partial E}{\partial t} + \nabla \cdot [(E + p)\mathbf{v}] = 0$$

You may see these equations rather a lot, and if you are not on the continuum stream you may not care about them, but they are my favourite equations so I keep using them

- From these, numerical methods for their solution are developed

$$u_i^{n+1} = u_i^n - \frac{\Delta t}{\Delta x}\left(f_{i+1/2}^{FORCE} - f_{i-1/2}^{FORCE}\right)$$

- And finally, we get to simulation results



- In order to derive, and understand the properties of these methods, we need appropriate mathematical techniques

Mathematical methods for scientific computing

- Throughout the MPhil in Scientific Computing, you will frequently need to use the mathematical techniques in this course — either for deriving or manipulating the equations you are going to solve, or for processing and understanding solution data
- Some of this course is likely to have been covered in previous courses you will have taken, but that may have been a while ago, or approached in a different way
- In this course, the aim is to set the methods lectured in the context of numerical methods and scientific computing
- Accompanying this course are a series of problems — these will help reinforce the content of this course, but will likely take longer than the gap between lectures, so can be completed throughout term, and for revision

Assumed knowledge — calculus

- Whilst this course will build on your previous maths background, it's worth repeating a few of the key expressions which will be used repeatedly, just in case they've faded from memory a little

1. The first fundamental theorem of calculus

$$\frac{d}{dx}\left(\int_{x_0}^{x} f(x')\,dx'\right) = f(x)$$

The derivative of the integral of a function returns the function itself (providing f is **continuous**)

2. The second fundamental theorem of calculus

$$\int_{x_0}^{x_1} \frac{df}{dx}\,dx = f(x_1) - f(x_0)$$

The integral of the derivative of a function, between two points, returns the difference of the function evaluated at these points (providing f is **differentiable**)

Assumed knowledge — series

3. Taylor series

$$f(x) = f(a) + \left.\frac{\partial f}{\partial x}\right|_{x=a}(x-a) + \left.\frac{1}{2}\frac{\partial^2 f}{\partial x^2}\right|_{x=a}(x-a)^2 + \left.\frac{1}{3!}\frac{\partial^3 f}{\partial x^3}\right|_{x=a}(x-a)^3 + \cdots$$

$$f(x+a) = f(x) + a\frac{\partial f}{\partial x} + \frac{a^2}{2!}\frac{\partial^2 f}{\partial x^2} + \frac{a^3}{3!}\frac{\partial^3 f}{\partial x^3} + \ldots$$

4. Fourier series for a function with period L

$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty}\left(a_n \cos\left(\frac{2\pi n x}{L}\right) + b_n \sin\left(\frac{2\pi n x}{L}\right)\right)$$

$$a_n = \frac{2}{L}\int_{x_0}^{x_0+L} f(x)\cos\left(\frac{2\pi n x}{L}\right)dx\,, \quad b_n = \frac{2}{L}\int_{x_0}^{x_0+L} f(x)\sin\left(\frac{2\pi n x}{L}\right)dx$$

Assumed knowledge — further identities

5. Binomial expansion, in particular

$$(1+x)^{-1} = 1 - x + x^2 - x^3 + \dots$$

6. Stokes' theorem

$$\oint_C \mathbf{F} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$$

7. Divergence theorem

$$\int (\nabla \cdot \mathbf{F})\, dV = \int_S \mathbf{F} \cdot d\mathbf{S}$$

Note — despite the last two often being core components of vector calculus, we won't really use these here: they appear more in analytic calculations than computational ones

Notation note

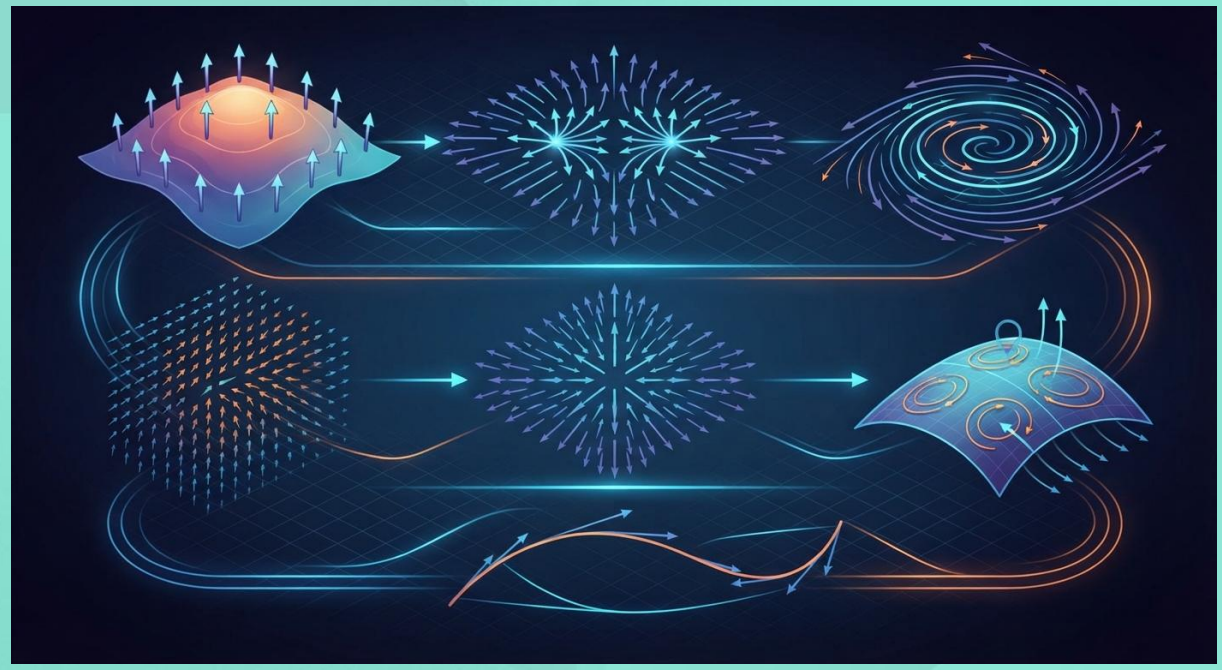
- In keeping with standard mathematical notation, vector quantities will always be emboldened, e.g. $\mathbf{a}$ is a vector, a is a scalar
- Often, vectors will be represented by lower case letters, matrices by upper case — but this is not universal
- The reason: conflicting conventions — for example, magnetic field is typically $\mathbf{B}$
- Matrices will be kept non-italic and non-bold (but I may forget, sometimes) — context is often more important than typeface
- When we move to index notation, indices are the only important thing to look at — they provide the context

Outline

- 1. Vector calculus
- 2. Index notation
- 3. Tensors

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Vector Calculus



Vector calculus and evolution equations

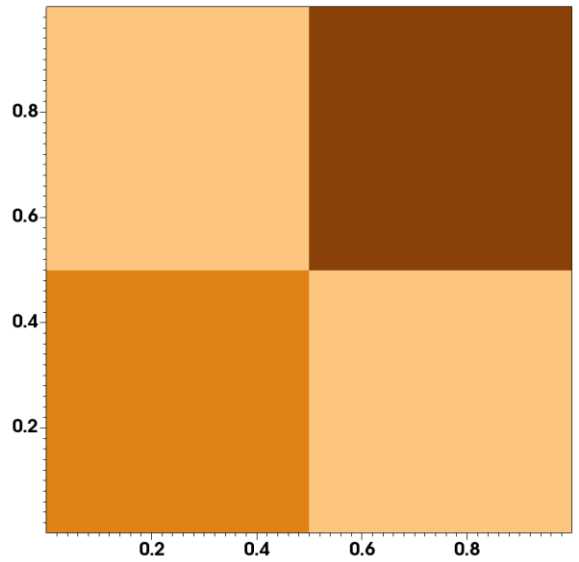
- The first topic we shall consider is vector calculus, with a key focus on index notation
- When describing physical systems, we expect to find both **vectors** (velocity, relative position) and **calculus** (gradients, rates of change)
- My favourite Euler equations are a good example of this:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$$

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} + Ip) = 0$$

$$\frac{\partial E}{\partial t} + \nabla \cdot [(E + p)\mathbf{v}] = 0$$

What are these equations?



- Those on the continuum stream will see these equations a lot, and those on the HPC stream probably can't avoid them either
- They were picked partly because I like them, but also because their natural form requires vector calculus operations
- If we want to rearrange such equations, we need vector calculus (or an excessive amount of paper by writing out every single component)
- Physically, these are just equations which describe the conservation of things we expect to be conserved:
  - Mass (or density, ρ)
  - Momentum ($\rho\mathbf{v}$)
  - Energy (E)
- The conservation of mass says, for example, that the change in mass over time is related to the gradient in momentum (a rate of flow of mass)

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$$

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} + Ip) = 0$$

$$\frac{\partial E}{\partial t} + \nabla \cdot [(E + p)\mathbf{v}] = 0$$

Vector form of equations

- Real-world simulations typically cannot be described by a single variable; we need to consider the interaction of multiple variables in a **coupled system**
- The density depends on the momentum, but the momentum itself also depends on density (and pressure)
- It is therefore common to find these systems described in “vector form”

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$$

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} + Ip) = 0$$

$$\frac{\partial E}{\partial t} + \nabla \cdot [(E + p)\mathbf{v}] = 0$$



$$\frac{\partial}{\partial t} \begin{pmatrix} \rho \\ \rho \mathbf{v} \\ E \end{pmatrix} + \nabla \cdot \begin{pmatrix} \rho \mathbf{v} \\ \rho \mathbf{v} \otimes \mathbf{v} + Ip \\ (E + p)\mathbf{v} \end{pmatrix} = 0$$

Unpacking this vector form

$$\frac{\partial}{\partial t}\begin{pmatrix}\rho\\ \rho\mathbf{v}\\ E\end{pmatrix}+\nabla\cdot\begin{pmatrix}\rho\mathbf{v}\\ \rho\mathbf{v}\otimes\mathbf{v}+Ip\\ (E+p)\mathbf{v}\end{pmatrix}=0$$

- Although a very concise definition of the equations, there is perhaps some unexpected notation
- We have the velocity, $\mathbf{v}$, within the vector brackets, i.e. we have placed vectors within vectors
- Really, we are abusing notation by having two forms of vectors; velocity is a mathematical vector, a quantity with direction and magnitude, whilst these brackets are more of a computational vector; a list of components – but this is scientific computing, so we are allowed to work from both worlds
- This is convenient here, because we can't write out the velocity terms explicitly in this list, since we haven't chosen our coordinate system yet
- Technically, the second vector is actually a matrix, and now we also have the divergence of a matrix
- This does have a reasonable mathematical definition — we shall come to this shortly

Coordinate-independent expressions

$$\frac{\partial}{\partial t}\begin{pmatrix}\rho\\ \rho\mathbf{v}\\ E\end{pmatrix}+\nabla\cdot\begin{pmatrix}\rho\mathbf{v}\\ \rho\mathbf{v}\otimes\mathbf{v}+Ip\\ (E+p)\mathbf{v}\end{pmatrix}=0$$

- It is common to reduce dimensionality by appealing to symmetries, e.g. cylindrical symmetry
- The vector-form equation holds regardless of whether

$$\mathbf{v}=v_x\mathbf{e}_i+v_y\mathbf{e}_j+v_z\mathbf{e}_k \quad \text{or} \quad \mathbf{v}=v_r\mathbf{e}_r+v_\theta\mathbf{e}_\theta+v_\varphi\mathbf{e}_\varphi$$

- Once you know your coordinate system, you might be able to simply use a form like

$$\begin{pmatrix}\rho\\ \rho\mathbf{v}\\ E\end{pmatrix}=\begin{pmatrix}\rho\\ \rho v_x\\ \rho v_y\\ \rho v_z\\ E\end{pmatrix}$$

- However, this coordinate-independent approach hides an additional complication: when expressing the equation in your chosen coordinates, the basis vectors themselves might have non-zero derivatives

Quick review of vector operations — gradient

- For now, we shall assume we are working in Cartesian coordinates, and define everything in terms of the three directions x, y and z
- In these coordinates, **gradient**, **divergence** and **curl** are the key calculus operators
- Gradient operates on scalar fields, and provides their rate of change in arbitrary directions as a vector field
- Note - $\nabla\alpha$ is perpendicular to the field α

$$\nabla\alpha = \begin{pmatrix} \dfrac{\partial\alpha}{\partial x} \\ \dfrac{\partial\alpha}{\partial y} \\ \dfrac{\partial\alpha}{\partial z} \end{pmatrix}$$

Quick review of vector operations — divergence

- Divergence (div) can be considered the rate at which a vector field enters or leaves a given volume

$$\nabla \cdot \mathbf{F} = \lim_{V \to 0} \frac{1}{V} \int_S \mathbf{F} \cdot \hat{\mathbf{n}} \, dS$$

$$\nabla \cdot \mathbf{a} = \frac{\partial a_x}{\partial x} + \frac{\partial a_y}{\partial y} + \frac{\partial a_z}{\partial z}$$

- This is why it appears in expressions like conservation of mass — it lets us track mass loss/gain in a volume
- Mathematically, this is technically **not** a dot product, but it is nearly universally used this way

Quick review of vector operations — curl

- Curl is a measure of circulation — related to the **rate of rotation**, not directly to the rotation itself
- It also has a definition as a limit of an integral

$$\nabla \times \mathbf{F} = \lim_{V \to 0} \frac{1}{V} \int_S \hat{\mathbf{n}} \times \mathbf{F} \, dS$$

$$\nabla \times \mathbf{a} = \begin{vmatrix} \mathbf{e}_i & \mathbf{e}_j & \mathbf{e}_k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ a_x & a_y & a_z \end{vmatrix} = \begin{pmatrix} \partial a_z/\partial y - \partial a_y/\partial z \\ \partial a_x/\partial z - \partial a_z/\partial x \\ \partial a_y/\partial x - \partial a_x/\partial y \end{pmatrix}$$

- Commonly encountered with electric and magnetic fields, since these act perpendicularly to each other

$$\frac{\partial \mathbf{B}}{\partial t} + \nabla \times \mathbf{E} = 0$$

- Again, technically this is not a true cross product mathematically

More vector operations — the outer product

- Sometimes less familiar operations appear — for example the outer product, $\otimes$, and the previously mentioned divergence of a matrix (tensor)
- The physical use of the outer product is, for example, when a quantity moving in one direction is affected by quantities in other directions
- E.g. a gradient in ρv_x can exist in the y-direction, so x-momentum can move in the y-direction
- Some papers are lazy, and write this as $\rho\mathbf{vv}$ (a non-dot-product multiplication of two vectors, column × row)

$$\mathbf{a} \otimes \mathbf{b} = \begin{pmatrix} a_x b_x & a_x b_y & a_x b_z \\ a_y b_x & a_y b_y & a_y b_z \\ a_z b_x & a_z b_y & a_z b_z \end{pmatrix}$$

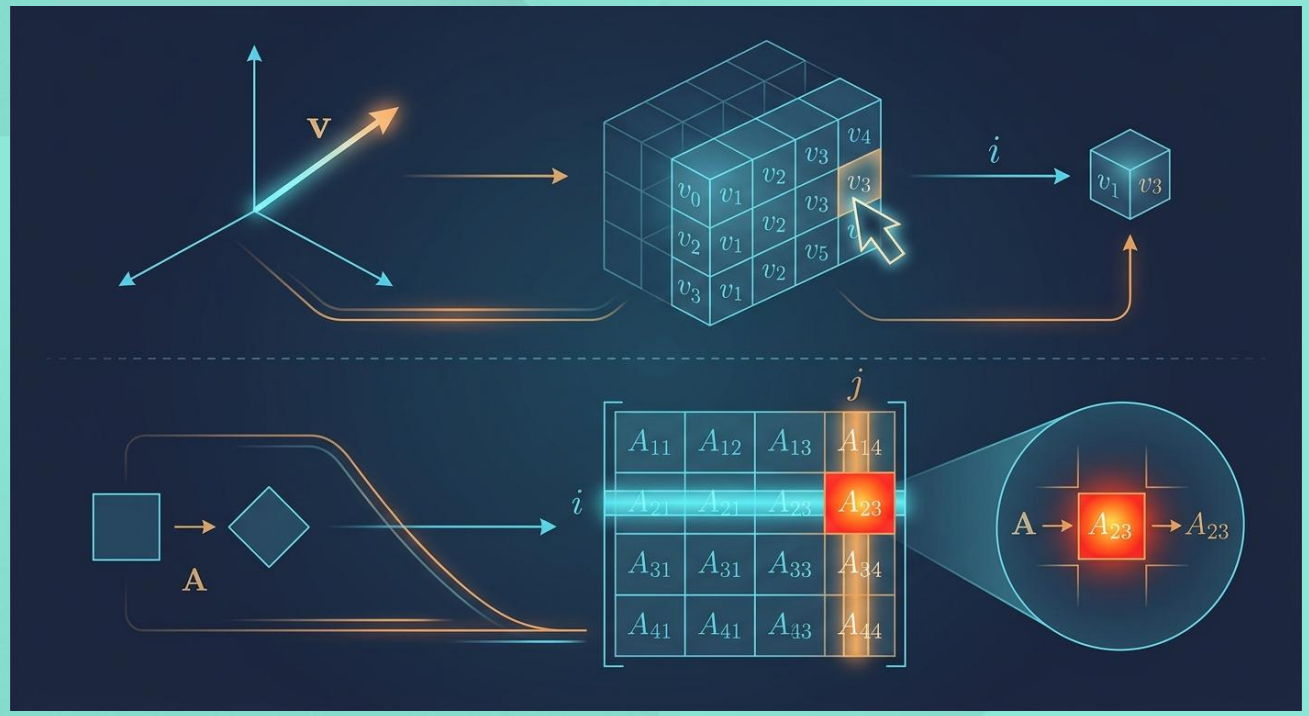
$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} + I p) = 0$$

More vector operations - the divergence of a matrix

$$\nabla \cdot \mathrm{A} = \begin{pmatrix} \partial A_{xx}/\partial x + \partial A_{xy}/\partial y + \partial A_{xz}/\partial z \\ \partial A_{yx}/\partial x + \partial A_{yy}/\partial y + \partial A_{yz}/\partial z \\ \partial A_{zx}/\partial x + \partial A_{zy}/\partial y + \partial A_{zz}/\partial z \end{pmatrix} \quad \text{or} \quad \nabla \cdot \mathrm{A} = \begin{pmatrix} \partial A_{xx}/\partial x + \partial A_{yx}/\partial y + \partial A_{zx}/\partial z \\ \partial A_{xy}/\partial x + \partial A_{yy}/\partial y + \partial A_{zy}/\partial z \\ \partial A_{xz}/\partial x + \partial A_{yz}/\partial y + \partial A_{zz}/\partial z \end{pmatrix} ?$$

- The divergence of a matrix is not actually a unique quantity, but the two versions are related to the divergence of a vector
- Treating it as a dot product, you either take a dot product along each row, or each column, of the matrix to get vector quantity
- At this point, you may ask whether there is a standard definition
- Unfortunately, the answer is no — you are often expected to know which version is in use
- Sometimes the first of these is called $\mathrm{div}\,\mathrm{A}$, and the second $\nabla \cdot \mathrm{A}$, but this is in no way universal, and most of the field of computational fluid dynamics doesn't do this
- Ideally, we need a less confusing way of defining these equations, one which we know will be unambiguous, which brings us to index notation

Index Notation



Index notation

- Some quantities are not natural to work with — for example, we may want to write $\nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v})$ as partial derivatives of just $\mathbf{v}$, without having to write down a separate expression for each dimension, or worse, for each matrix entry
- Additionally, for some cases (such as elasticity) we end up with rank-4 tensors (like a matrix, but in four dimensions...)
- In such cases, **index notation** makes life much more straightforward
- For example, this vector equation (where τ is a matrix)

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} + Ip) = \nabla \cdot \tau + \rho \mathbf{f}$$

becomes

$$\frac{\partial \rho v_i}{\partial t} + \nabla_j (\rho v_i v_j + \delta_{ij} p) = \nabla_j \tau_{ij} + \rho f_i$$

Reading index notation — scalars and vectors

- The (unique) indices tell you what you are dealing with
- In this expression, i is a unique index, it appears only once in each term, (ignore j for now which appears either twice or not at all)
- Scalars have **no indices**, e.g. ρ and p
- Vectors have **one index**
- In Cartesian coordinates, the (1,2,3) and (x, y, z) subscripts are often used to mean the same thing, $(v_1, v_2, v_3)^T = (v_x, v_y, v_z)^T$

$$\frac{\partial \rho v_i}{\partial t} + \nabla_j(\rho v_i v_j + \delta_{ij} p) = \nabla_j \tau_{ij} + \rho f_i$$

$$\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \rightarrow v_i \mathbf{e}_i \text{ or } v_i, \qquad \mathbf{v} = (v_x, v_y, v_z)^T$$

- Note — $v_i \mathbf{e}_i$ could be considered partial index notation: vector components are represented as indices, but operators (e.g. dot products) aren't, and is a simple way of writing $\mathbf{v} = v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2 + v_3 \mathbf{e}_3$, the typical way of writing a vector in terms of basis vectors

Reading index notation — matrices

$$\frac{\partial \rho v_i}{\partial t} + \nabla_j(\rho v_i v_j + \delta_{ij} p) = \nabla_j \tau_{ij} + \rho f_i$$

- Matrices have two indices: the first gives the row, the second gives the column

$$\tau = \begin{pmatrix} \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} \end{pmatrix} = \begin{pmatrix} \tau_{11} & \tau_{12} & \tau_{13} \\ \tau_{21} & \tau_{22} & \tau_{23} \\ \tau_{31} & \tau_{32} & \tau_{33} \end{pmatrix} \rightarrow \tau_{ij}$$

$$I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \delta_{11} & \delta_{12} & \delta_{13} \\ \delta_{21} & \delta_{22} & \delta_{23} \\ \delta_{31} & \delta_{32} & \delta_{33} \end{pmatrix} \rightarrow \delta_{ij}$$

- The identity matrix, written in index notation, is typically called the **Kronecker delta**

Combined quantities

$$\frac{\partial \rho v_i}{\partial t} + \nabla_j(\rho v_i v_j + \delta_{ij} p) = \nabla_j \tau_{ij} + \rho f_i$$

- Everything separated by addition, subtraction or an equals sign **must** be the same rank — i.e. the same type of thing
- Probably an obvious statement, when considering our starting point for these equations, since this is equivalent to statements such as “you can't add a vector to a matrix"
- Similarly, everything within a set of parentheses (or within an operator) must locally also be the same thing
- So here, both $\rho v_\mathrm{i} v_\mathrm{j}$ and $\delta_\mathrm{ij} p$ have two indices, and hence are both matrices
- $\delta_\mathrm{ij} p$ is a scalar multiplied by a matrix, so we expect a matrix
- As is the outer product quantity, $\rho v_i v_j$, it is now very clear to see what the entries of this matrix should be

Free indices

$$\frac{\partial \rho v_i}{\partial t} + \nabla_j(\rho v_i v_j + \delta_{ij} p) = \nabla_j \tau_{ij} + \rho f_i$$

- We have so far avoided talking about what the $\nabla_j \tau_{ij}$ term is, (or how the ∇_j affects the matrix inside the brackets)
- It should be a vector, since $\frac{\partial \rho v_i}{\partial t}$ and ρf_i are vectors
- It also has one index repeated (j) and one which only appears once (i)
- A **free index** (previously called a unique index) is one that appears only once per term and this is what determines what type of quantity you have
- Repeated indices represent a summation

Repeated indices

- Repeated indices represent a sum over that index — the Einstein summation convention

$$v_i v_i = \sum_{i=1}^{3} v_i v_i = v_1 v_1 + v_2 v_2 + v_3 v_3 = \mathbf{v} \cdot \mathbf{v} = v^2$$

- Used for simplifying notation – we can end up with a lot of repeated indices in expressions, and writing out the sum every time (especially for nested summation) gets tiring
- This index can be repeated in two different quantities (such as the dot product above), or within a single quantity, as in Hooke's law for stress and strain

$$\sigma_{ij} = \lambda \delta_{ij} e_{kk} + 2\mu e_{ij}, \qquad e_{kk} = e_{11} + e_{22} + e_{33} = \mathrm{Tr}(e)$$

Dummy indices

- For free indices, the position of the index letter must be used consistently — for example these are all different quantities

$$C_{ij} = A_{ij} + B_{ij}, \qquad \hat{C}_{ij} = A_{ji} + B_{ji}, \qquad \bar{C} = A_{ij} + B_{ji}$$

Dummy indices

- For free indices, the position of the index letter must be used consistently — for example these are all different quantities

$$C_{ij} = A_{ij} + B_{ij}, \qquad \hat{C}_{ij} = A_{ji} + B_{ji}, \qquad \bar{C} = A_{ij} + B_{ji}$$

- However, if you have lots of sums, it doesn't matter what letter you actually use for the sum

$$v_i v_i = \sum_{i=1}^{3} v_i v_i = v_1 v_1 + v_2 v_2 + v_3 v_3 = \sum_{j=1}^{3} v_j v_j = v_j v_j$$

- As a result, repeated indices are sometimes known as **dummy indices**, because they can be freely renamed whenever you like
- As a result, we frequently rename repeated indices to make calculations easier to read – don't be alarmed if a_{ii} becomes a_{jj} in a subsequent line of a calculation – it is the same thing

Vector and matrix multiplication

- We have sort of introduced this already with the dot product and the summation convention
- But, unsurprisingly, since vector-matrix and matrix-matrix multiplication involve summation over rows or columns, so these can be described using index notation too
- For example, since

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} a_{11}b_1 + a_{12}b_2 \\ a_{21}b_1 + a_{22}b_2 \end{pmatrix} = \begin{pmatrix} \sum_j a_{1j}b_j \\ \sum_j a_{2j}b_j \end{pmatrix}$$

  we can easily see this can be written

$$\mathrm{A}\mathbf{b} \rightarrow A_{ij}b_j,$$

- Similarly

$$\mathrm{AB} \rightarrow A_{ij}B_{jk}$$

Indices — summary

- v_i, ε_{iij}, $a_{ij}b_j$: vector (one free index)
- τ_{ij}, $v_i v_j$, $a_{ij}b_{jk}$: matrix (two free indices)
- τ_{ii}: scalar (no free indices)
- τ_{11}: a specific entry of the matrix $\tau_{ij}, i = 1, j = 1$
- C_{ijkl}: rank-four object (e.g. the elasticity tensor)
- C_{iikl}: matrix (two free indices)
- C_{iiim}, ε_{iii}, $a_i + B_{ij}$: ❓ ❓ ❓ something has gone wrong – check your calculations!

Back to grad, div and curl

- The grad operator, ∇, is a vector operator, so can be written with index notation

$$\nabla = \begin{pmatrix} \partial/\partial x \\ \partial/\partial y \\ \partial/\partial z \end{pmatrix} \rightarrow \nabla_i = \frac{\partial}{\partial x_i},$$

- Grad and div are now straightforward

$$\nabla\alpha \rightarrow \nabla_i \alpha, \quad \nabla \cdot \mathbf{a} \rightarrow \nabla_i a_i$$

- Curl requires the addition of a new quantity, the **Levi-Civita symbol**

$$\nabla \times \mathbf{a} \rightarrow \epsilon_{ijk} \nabla_j a_k, \quad \epsilon_{ijk} = \begin{cases} +1 & (i,j,k) = (1,2,3), (2,3,1), (3,1,2) \\ -1 & (i,j,k) = (3,2,1), (1,3,2), (2,1,3) \\ 0 & i = j, \ j = k, \ \text{or} \ k = i \end{cases}$$

The Levi-Civita symbol

$$\nabla \times \mathbf{a} = \epsilon_{ijk} \nabla_j a_k$$

- There is one free index in $\varepsilon_{ijk} \nabla_j a_k$, so, as expected, this is a vector term
- The term $\nabla_j a_k$ is a matrix, and the Levi-Civita symbol a '3-D matrix'

$$\begin{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \end{pmatrix}$$

$$\nabla_j a_k = \begin{pmatrix} \partial a_x/\partial x & \partial a_y/\partial x & \partial a_z/\partial x \\ \partial a_x/\partial y & \partial a_y/\partial y & \partial a_z/\partial y \\ \partial a_x/\partial z & \partial a_y/\partial z & \partial a_z/\partial z \end{pmatrix}$$

- The quantity $\varepsilon_{ijk} \nabla_j a_k$ then multiplies each 'slice' of ε_{ijk} with $\nabla_j a_k$
- Also note that any cross product is written in index notation with the Levi-Civita symbol,

$$\mathbf{a} \times \mathbf{b} = \epsilon_{ijk} a_j b_k$$

Vector calculus in index notation

- Working in index notation has a few advantages
- Notation for matrices, and ‘higher dimensional matrices’, is straightforward, and there is no other universal convention for these quantities
- Simplification of expressions is also more straightforward — there is no need to memorise any relationships
- Well, maybe one:

$$\epsilon_{ijk}\epsilon_{ilm} = \delta_{jl}\delta_{km} - \delta_{jm}\delta_{kl}$$

- Dealing with derivatives is a lot easier too
- This is all because in index notation you are treating everything at the level of a scalar quantity, which means you can re-order things mid-calculation without having to worry about what this quantity actually is, or whether it would make sense written down (we shall see some examples of this soon)

Changing indices with the Kronecker delta

- One of the key features of vector calculus operations in index notation is using the Kronecker delta to change indices
- Sometimes this is obvious — multiplying a vector by the identity matrix does not change the vector

$$\delta_{ij} a_j = a_i$$

- Sometimes it takes a bit more thought

$$\delta_{ij} A_{ij} = A_{ii}, \qquad \delta_{im}\delta_{jn} B_{mnkl} = B_{ijkl}$$

- And occasionally, we work backwards

$$\nabla_j \rho v_i v_j + \nabla_i p = \nabla_j (\rho v_i v_j + \delta_{ij} p)$$

Some examples

- All ‘classic’ vector calculus identities can be demonstrated using index notation

$$\nabla \cdot (\mathbf{a} \times \mathbf{b}) = (\nabla \times \mathbf{a}) \cdot \mathbf{b} - \mathbf{a} \cdot (\nabla \times \mathbf{b})$$

Some examples

- All ‘classic’ vector calculus identities can be demonstrated using index notation

$$\nabla \cdot (\mathbf{a} \times \mathbf{b}) = (\nabla \times \mathbf{a}) \cdot \mathbf{b} - \mathbf{a} \cdot (\nabla \times \mathbf{b})$$

- Throughout the continuum courses, we will use index notation for expressing equations in ‘desired’ forms
- For example, if we want to solve the Maxwell–Faraday equation (with a few other coupled equations, and using Ohm's law),

$$\frac{\partial \mathbf{B}}{\partial t} + \nabla \times \mathbf{E} = 0, \qquad \eta \mathbf{J} = \mathbf{E} + \mathbf{v} \times \mathbf{B}, \qquad \eta = 0$$

then it is often desirable to express all spatial derivatives through a divergence operator

$$\frac{\partial \mathbf{B}}{\partial t} + \nabla \cdot (\mathbf{B} \otimes \mathbf{v} - \mathbf{v} \otimes \mathbf{B}) = 0$$

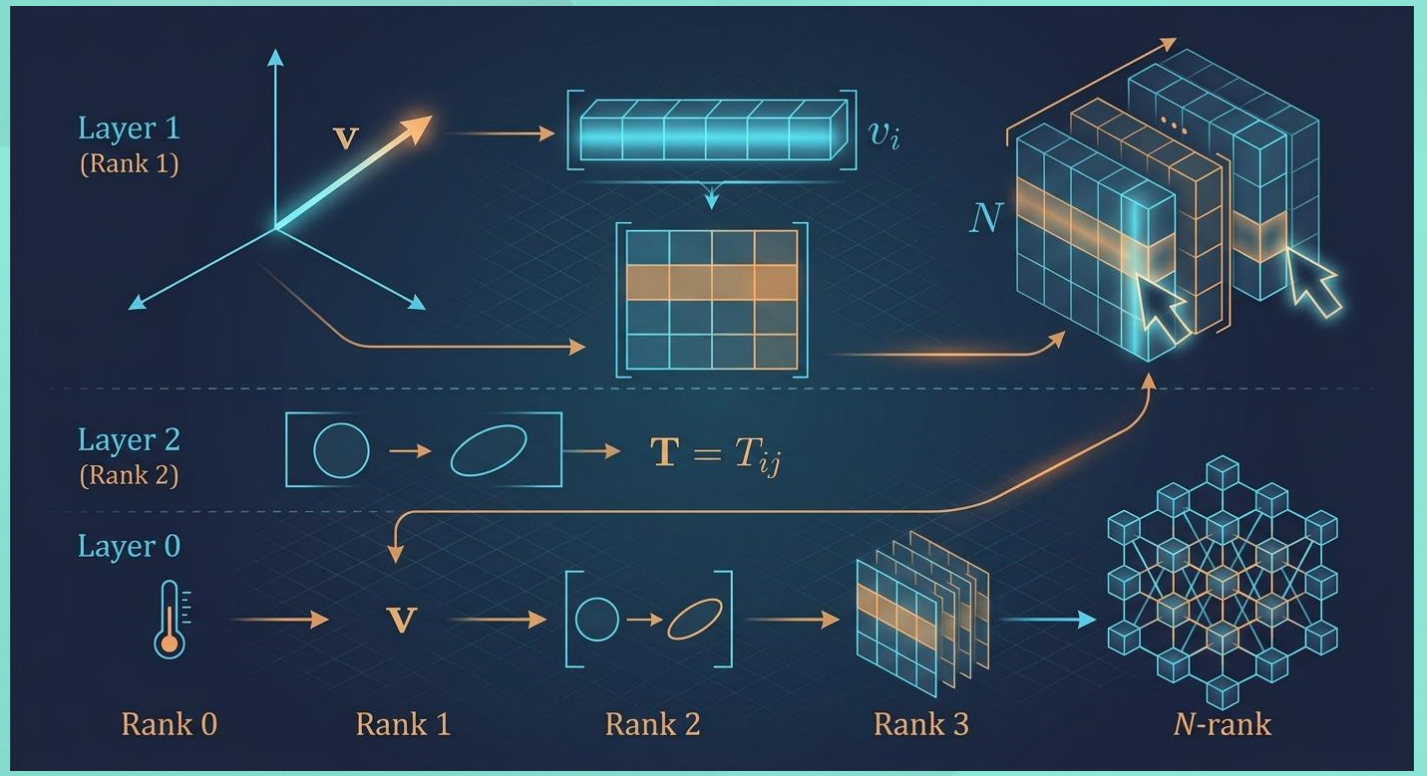
A disclaimer

- Technically, index notation does not just use subscripts, but also superscripts

$$a^i = \begin{pmatrix} a^x \\ a^y \\ a^z \end{pmatrix}, \qquad a_i = (a_x, a_y, a_z)$$

- Because we tend to focus on Cartesian coordinates, we ignore this distinction (we have $a^x = a_x$ and so on)
- But in more general coordinate systems this is not true, and a transformation is required to move between the two forms
- We rarely encounter this in this MPhil; we even write non-Cartesian coordinates in a ‘Cartesian-like’ manner
- Fusion plasma equilibrium literature is one notable exception — non-orthogonal coordinate systems are common in much of the traditional literature

Tensors


Layer 1
(Rank 1)
v
v_i
N
Layer 2
(Rank 2)
$\mathbf{T} = T_{ij}$
Layer 0
v
Rank 0
Rank 1
Rank 2
Rank 3
N-rank

Tensors

- Using index notation makes it easy to write down a 'three-dimensional matrix' in a straightforward way
- It is common to refer to these sorts of quantities as **tensors**, though in reality, tensors have a slightly stricter definition
- This starts with vectors, which are expressed in a given coordinate system through **basis vectors**

$$\mathbf{a} = a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + a_3\mathbf{e}_3$$

$$\mathbf{a} = a'_1\mathbf{e}'_1 + a'_2\mathbf{e}'_2 + a'_3\mathbf{e}'_3$$

- For example, we may have Cartesian basis vectors $\mathbf{e}_x$, $\mathbf{e}_y$, $\mathbf{e}_z$, (often written $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$), or spherical polar basis vectors $\mathbf{e}_r$, $\mathbf{e}_\phi$, $\mathbf{e}_\theta$
- Coordinates transform through

$$a_i = \frac{\partial x_i}{\partial x'_j} a'_j \quad \text{or} \quad a_i = R_{ij}a'_j \quad \text{or} \quad a_i = L_{ij}a'_j$$

Jacobian matrix, or just Jacobian

Note – these last two typically correspond specifically to rotation or linear transformation of a coordinate system

Why do we care about tensors?

- Vectors can always be transformed through the Jacobian
- A tensor is a generalisation of this – any quantity that also transforms according to these transformation rules, e.g.

$$t_{ij} = R_{ik}R_{jl}t'_{kl} \qquad \text{or} \qquad t_{ijkl} = R_{im}R_{jn}R_{kp}R_{lq}t'_{mnpq}$$

- The number of indices is also known as its **rank**
- One of the key aspects of a tensor is that the quantity represents the same physical thing, **no matter what coordinate system we are working in** (though individual entries will, of course, be different)
- This is important for consistent definitions of quantities such as stress, or elastic deformation
- Though perhaps easiest to understand through quantities such as speed, which is the magnitude of velocity; if computing the magnitude of a velocity in different coordinate systems gave different answers, it would not be a tensor (but, fortunately, this is not the case)
- We see this importance in **invariant quantities**

Invariant quantities

- Invariant quantities are those which are the same, no matter the coordinate system used to compute them
- For a rank-2 tensor, the invariant quantities come from finding its eigenvalues (if you've forgotten what eigenvalues are, we will cover them in much more detail soon!)

$$\begin{vmatrix} a_{11}-\lambda & a_{12} & a_{13} \\ a_{21} & a_{22}-\lambda & a_{23} \\ a_{31} & a_{32} & a_{33}-\lambda \end{vmatrix} = 0$$

- This determinant can be written as a cubic equation with three coefficients $\lambda^3 - I_1\lambda^2 + I_2\lambda - I_3 = 0$

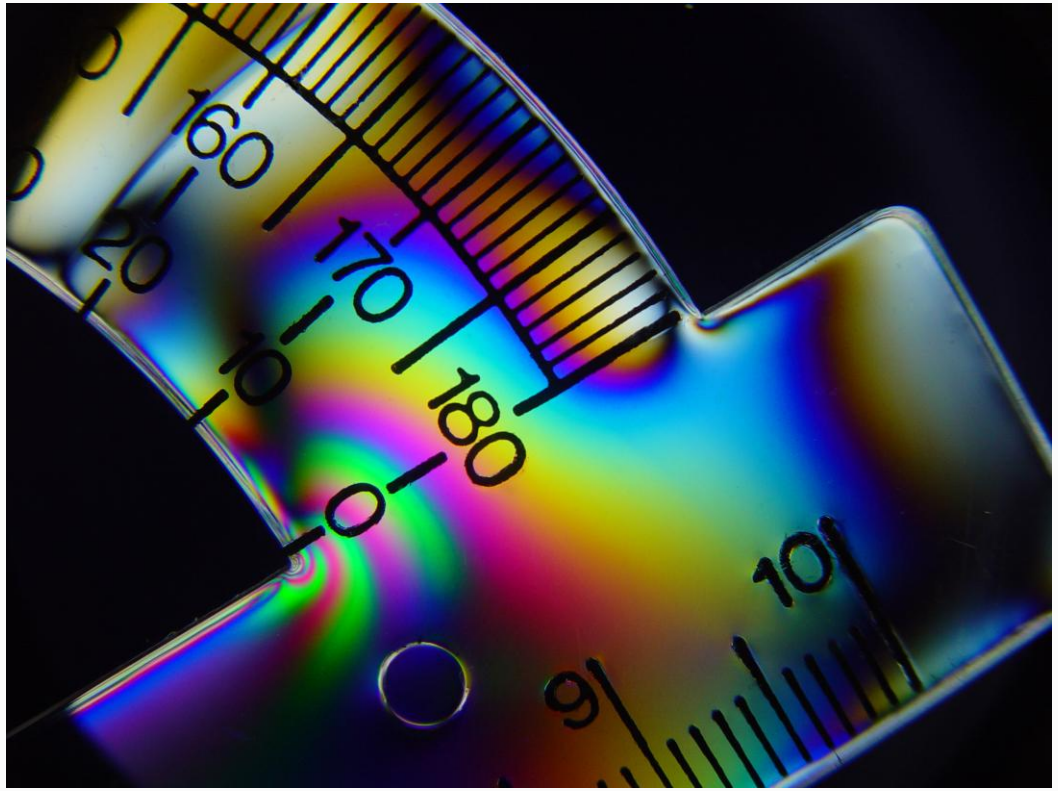
$$I_1 = a_{ii} = a_{11} + a_{22} + a_{33} = \mathrm{tr(A)} \qquad I_2 = \frac{1}{2}\left(a_{jj} - a_{ij}a_{ji}\right) = \frac{1}{2}\left[\left(\mathrm{tr(A)}^2 - tr(\mathrm{A}^2)\right)\right] \qquad I_3 = \det|\mathrm{A}|$$

- I_1, I_2 and I_3 are the **invariants** of a rank-2 tensor in 3D

Invariants, pressure and stress

- When modelling any continuum material, stress (σ_{ij}) is a rank-2 tensor which describes how the material is being pushed (or pulled) in any direction
- Pressure (p) is a scalar quantity, which gives some information about how much stress the material is under
- Perhaps unsurprisingly, the pressure experienced at any point within a material does not depend on the coordinate system used to calculate it — it is an invariant quantity

$$p = \frac{1}{3} I_1 = \frac{1}{3} \mathrm{tr}(\sigma)$$



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Matrices which aren't tensors

- Every rank-2 tensor is a matrix, but not every matrix is a tensor
- Trivially, any matrix given just as numbers, but without a coordinate system, will not be a tensor
- More importantly, though, there are two commonly encountered quantities which are matrices, relate obviously to coordinate directions, but are not tensors
- The first is the coordinate transformation, or Jacobian

$$\mathbf{e} = \frac{\partial \mathbf{x}}{\partial \mathbf{x}'}\mathbf{e}' \quad \text{or} \quad e_i = R_{ij}e'_j$$

- The second is the **deformation gradient**, F_{ij} — this transforms an infinitesimal particle from its reference configuration $\mathrm{d}\mathbf{X}$ to its current position $\mathrm{d}\mathbf{x}$, by deformation (squashing and expanding) alone (no rotation or translation)

$$F_{ij} = \frac{dx_i}{dX_i}$$

What are these matrices?

- Both these examples take one vector ($\mathbf{e}'$ or $\mathbf{X}$) and transform it to a new vector ($\mathbf{e}$ or $\mathbf{x}$)
- These quantities **transform as vectors** — i.e. would need one coordinate transform, whilst a regular rank-2 tensor would require two
- On inspection, this is perhaps not surprising: when we perform a coordinate transformation, this would either be in the coordinates we started with or with the coordinates we end up with
- Such quantities are known as **two-point tensors** (which, despite still having tensor in this name, does not make it a tensor), or **double vectors**
- This name does not always make it to the literature, though...